

CASE STUDY

3D-PRINTED CONCRETE FROM WASTE

Multi-objective optimization tuned the mix and the printer together, replacing up to **97% of natural sand** with recycled concrete while improving buildability by **66%**, so walls printed **6 to 10 layers** high without collapse.

Printing Structural Concrete With Recycled Sand

3D concrete printing promises faster, safer construction with less labour and more design freedom. Making it sustainable means replacing natural sand with construction and demolition waste (CDW), but recycled sand absorbs more water, varies from batch to batch and is harder to extrude. This work, carried out at the University of Arizona, shows how computational intelligence can tune both the material and the machine so that high-CDW concrete still prints cleanly and holds its shape.

OVERVIEW

A printable mix has to flow through the pump and nozzle, then stiffen fast enough to carry the layers above it. Change the recycled content and every setting shifts: water, admixture, silica fume, print speed and pump rate.

Instead of tuning these one at a time, a multi-objective Bayesian optimizer searched mixture design and printing parameters together, chasing two goals at once: more recycled content and better buildability.

THE PROJECT

Recycled concrete was crushed, milled and sieved below 2 mm to suit extrusion, then blended with Portland cement, silica fume, natural sand, water and a high-range water reducer.

Mixes were printed with an industrial robotic arm through a 20 mm nozzle, building a 200 × 350 × 400 mm test wall. Buildability was scored as the number of layers printed before collapse.

At a Glance

- **Feedstock:** recycled concrete sand below 2 mm, replacing 52% to 97% of natural sand.
- **Binders:** Type II/V Portland cement with 1% to 3% silica fume.
- **Goals:** maximise recycled content and buildability at the same time.
- **Method:** Gaussian process model with a hypervolume-based multi-objective search.
- **Validation:** fresh-state tests, robotic printing, strength in three directions, XRD and SEM.



Maximise Recycled Content

Push CDW as far as possible as a replacement for natural sand.



Build Higher

Print more layers without slumping, cracking or collapse.



Tune Mix and Machine

Optimise ingredients, print speed and pump rate in one loop.

The Challenge

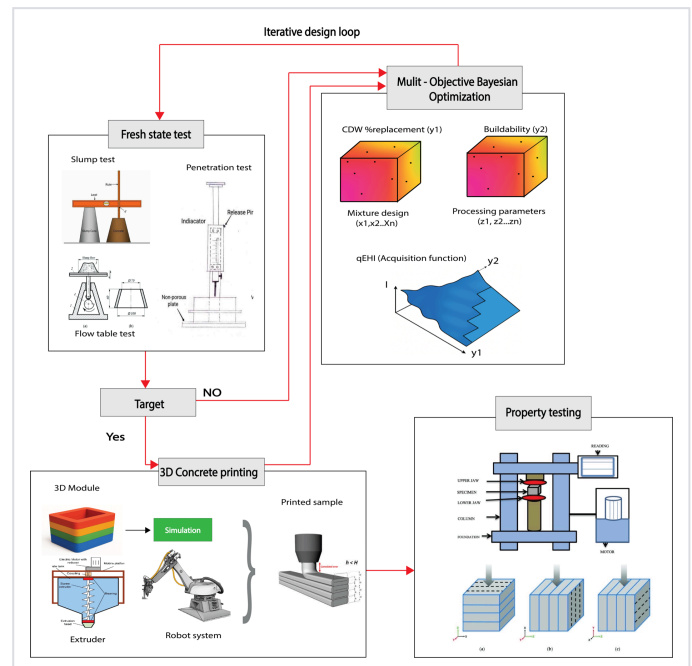
Recycled concrete sand absorbs about four times more water than natural sand (9.76% vs. 2.34%) and its particles vary in size and shape. In a printer that means unstable flow, broken filaments and layers that sag under their own weight.

Mixture and process are tightly linked: if the nozzle's travel speed and the pump's flow rate fall out of step, the filament thins or tears. Starting mixes held only 30% to 50% recycled content and supported just one to five layers.



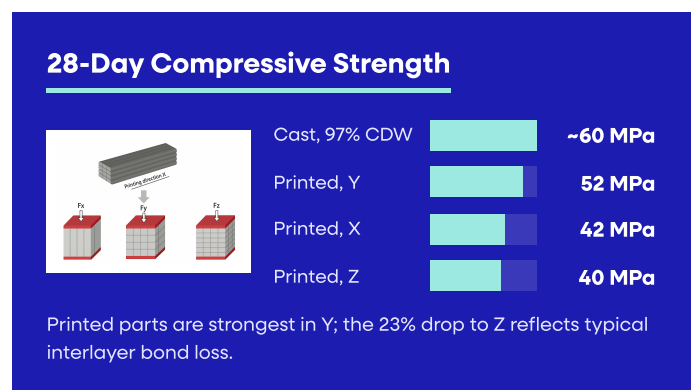
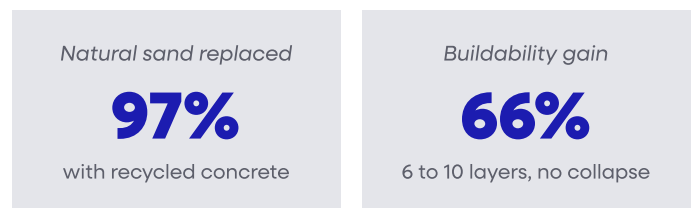
How It Works

Every round runs the same loop: fresh-state tests screen each mix, printable mixes are built and scored, and the results feed the optimizer, which proposes the next mixes and print settings to try.



Results

Guided by the optimizer, recycled content rose from 30%–50% to between 52% and 97%, while buildability improved by 66%. At a print speed of 100 mm/s and pump rates of 150 to 190 rpm, the 97%, 80% and 52% CDW mixes printed 6, 9 and 10 layers, with wall dimensions within ± 3 mm of the design.



Results in Detail

Three winning mixes were carried forward: CDWSF-3 (52% CDW, 2.40% silica fume), CDWSF-2 (80% CDW, 2.40%) and CDWSF-1 (97% CDW, 2.55%). Here is how the search found them and how they perform.

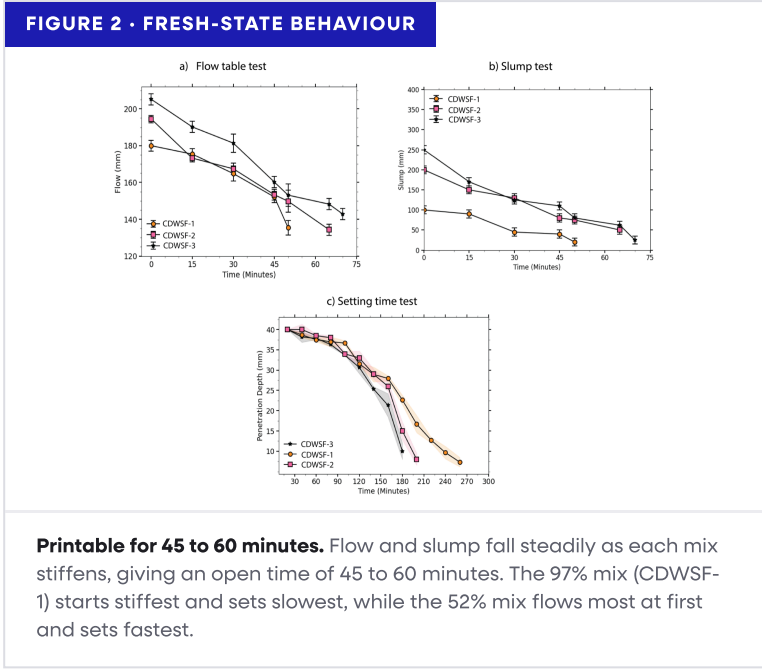
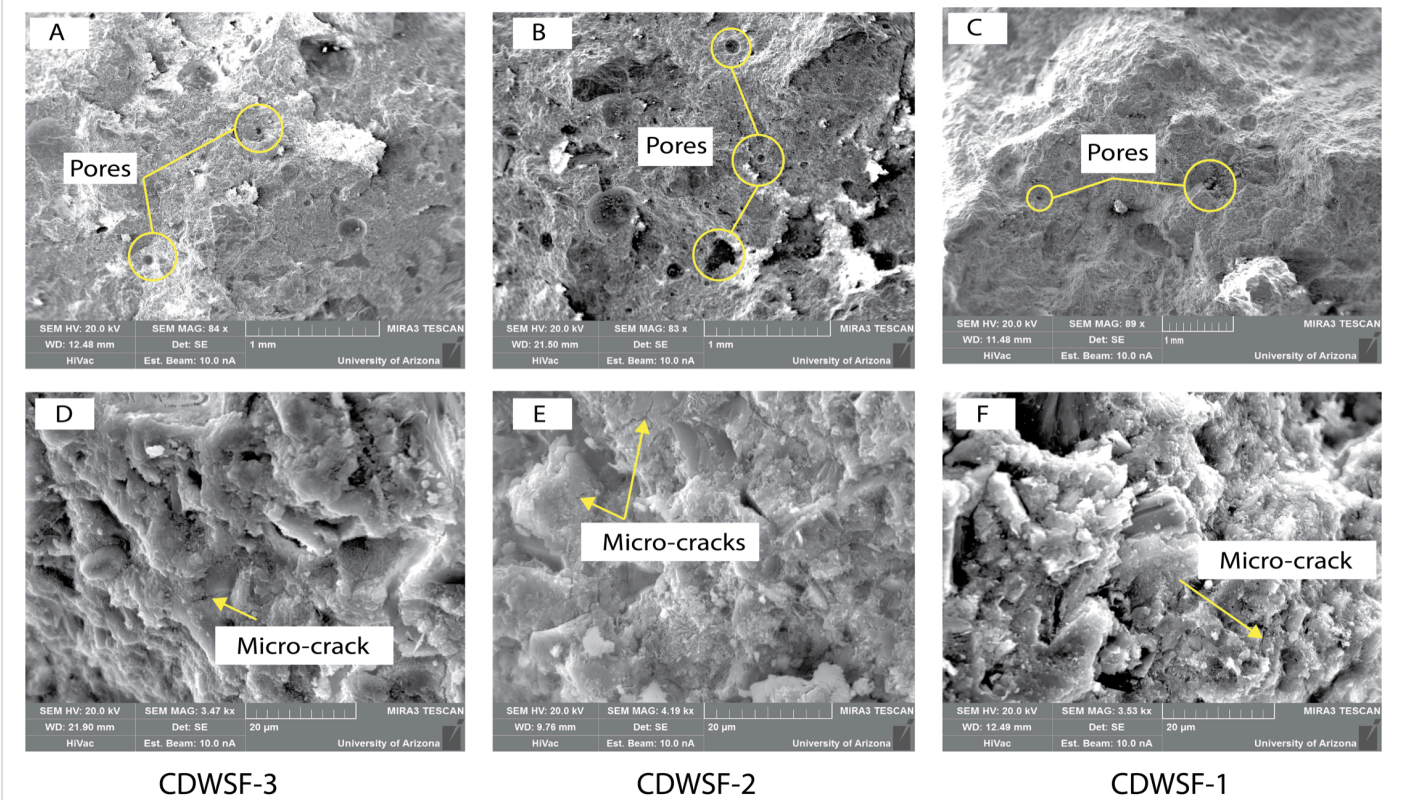


FIGURE 3 · MICROSTRUCTURE (SEM, 28 DAYS)



A denser matrix. Electron microscope images of CDWSF-3, CDWSF-2 and CDWSF-1 at low (top) and high (bottom) magnification show pores and fine micro-cracks. XRD confirmed that silica fume consumed portlandite to form extra binding gel (C-S-H), densifying the high-CDW mixes.